

THE ZERO PARADOX

ZP-H Addendum

The Snap Floor in Native Categories

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ZP-H derives the Binary Snap across four mathematical frameworks and locates the snap floor $\perp$ as the categorical bottom in each. In the main document that location is carried by three depth-index proxy categories — copies of $\mathbb{N}$ with the order $\leq$ ($Q_2\text{BallDepth}$, InfoDepth , HilbDimDepth), engineered so that depth 0 is the initial object. This addendum does the stronger thing: it drops the snap floor into each framework's own category, taken directly from Mathlib, where the floor appears in that field's native syntax. A topologist, a functional analyst, and a probabilist each meet the same bottom, each in their own category and their own language.

Three genuine Lean functors realize the snap chain inside the standard categories: F_B into TopCat (topological spaces), F_D into $\text{ModuleCat } \mathbb{C}$ ($\mathbb{C}$ -modules), and F_C into the Kleisli category of the probability monad PMF (stochastic maps). Each is sorry-free, and each carries the snap floor to its category's categorical bottom — an inverse limit in TopCat , an initial object in $\text{ModuleCat } \mathbb{C}$ and in the Kleisli category. The bundled witness is `mc1_correspondence` (`MC1Bridge.lean`).

Two honesty boundaries hold throughout. The realized claim is correspondence, not identity. What is proved is that each bottom is its own category's categorical bottom, agreeing on the snap; that the four bottoms are numerically one object is retired as ill-typed (object equality across categories does not typecheck and is not invariant under equivalence); what separates the members is proved property by property (`seam_unique_among_named`, for the named bottoms, in a lattice with no top). The native categories are the honest target. Mathlib has no bespoke category of p-adic spaces, Hilbert spaces, or information spaces, so the general-purpose categories — topological spaces, $\mathbb{C}$ -modules, stochastic maps — are where the floor genuinely lives.

Section I: From Depth Proxies to Native Categories

The proxy categories of the main document are correct but indirect. $Q_2\text{BallDepth}$, InfoDepth , and HilbDimDepth are each a copy of $\mathbb{N}$ with the order $\leq$, wrapped so that the bottom 0 satisfies the universal property of an initial object and the source-asymmetry condition AX-G2 (no morphism returns to 0). They witness the snap floor's categorical role, but inside a structure built for the purpose rather than inside the domain itself.

The native categories are not built for the purpose, and that is the point — they also do not satisfy the ZP categorical axioms. A ZPCategory (ZP-G) requires that no terminal object exist (AX-G1): the snap has nowhere higher to go. TopCat and $\text{ModuleCat } \mathbb{C}$ both have a terminal object (the one-point space; the zero module). So the snap floor cannot be "the initial object of a ZPCategory " here. The honest statement is the

one each category supports natively: in TopCat the floor is the limit of the shrinking system, and in ModuleCat $\mathbb{C}$ and the Kleisli category it is the genuine initial object. Each is stated below.

Section II: F_B — The Snap Floor in TopCat

The 2-adic snap floor is $0 \in \mathbb{Q}_2$. Around it sits the descending tower of clopen balls $B(0, 2^{-n})$ — the unit ball, then radius 1/2, then 1/4, and so on — each a genuine topological subspace of $\mathbb{Q}_2$. The inclusions $B(0, 2^{-m}) \subseteq B(0, 2^{-n})$ for $n \leq m$ are continuous, so the tower is a functor out of $\mathbb{N}^{op}$ (the balls shrink as the index grows): a real diagram in TopCat.

Functor: <code>fB_functor : $\mathbb{N}^{op} \rightarrow \text{TopCat}$ (TopFunctor.lean)</code>
Object $n \mapsto$ the clopen ball $B(0, 2^{-n}) \subseteq \mathbb{Q}_2$, as a TopCat object.
Morphism ($n \leq m$): the continuous inclusion of the smaller ball into the larger.
<code>fB_bottom_is_limit</code> : $\bigcap_n B(0, 2^{-n}) = \{0\}$
The snap floor is the inverse limit of the tower — the one point common to every ball.
Sorry-free. Purity [<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>].

TopCat has both a terminal object (the one-point space) and an initial object (the empty space); the snap floor is neither. It is the limit of the ball tower — the categorical expression a topologist would expect for "the point everything converges to." This is the honest realization in a category that is not a ZPCategory: not an initial object, but a limit.

Section III: F_D — The Snap Floor in ModuleCat $\mathbb{C}$

The state layer represents the snap in a complex Hilbert space: StateSpace n is the n -dimensional space $\mathbb{C}^n$, with the snap an orthogonal shift between basis states (ZP-D). As $\mathbb{C}$ -modules, these spaces and the linear maps between them form a diagram in ModuleCat $\mathbb{C}$.

Functor: <code>fD_functor : $\mathbb{N} \rightarrow \text{ModuleCat } \mathbb{C}$ (HilbFunctor.lean)</code>
Object $n \mapsto$ StateSpace n = the $\mathbb{C}$ -module $\mathbb{C}^n$.
Morphism ($n \leq m$): the isometric $\mathbb{C}$ -linear embedding that pads the extra coordinates with zero.
<code>fD_zero_isInitial</code> : StateSpace 0 is the initial object of ModuleCat $\mathbb{C}$.
The snap floor is the zero module — the one-element $\mathbb{C}$ -module $\{0\}$, which is the categorical zero object (both initial and terminal).
<code>fD_embed_inner</code> : the embeddings preserve the inner product (genuine isometries).

Functor: $\mathbf{fD_functor} : \mathbb{N} \rightarrow \mathbf{ModuleCat} \ \mathbb{C} \ (\mathbf{HilbFunctor.lean})$

Sorry-free. Purity [propext, Classical.choice, Quot.sound].

Here, unlike TopCat, the floor is the initial object outright: the zero object of $\mathbf{ModuleCat} \ \mathbb{C}$ is $\mathbf{StateSpace} \ 0$. The isometry lemma records what $\mathbf{ModuleCat} \ \mathbb{C}$ forgets — that the embeddings are not merely linear but inner-product preserving — so the Hilbert-space structure travels alongside the functor rather than being discarded by the choice of category.

Section IV: $\mathbf{F_C}$ — The Snap Floor in KleisliCat PMF

The information layer measures the snap as one bit of divergence between distributions ($\mathbf{ZP-C}$, $\mathbf{JSD} = \log 2$). Its native category is the Kleisli category of the finite-distribution monad PMF: objects are finite types, and a morphism $A \rightarrow B$ is a stochastic map (a Markov kernel), assigning to each $a \in A$ a probability distribution on B . Mathlib supplies this category for free, since PMF is a lawful monad.

Functor: $\mathbf{fC_functor} : \mathbb{N} \rightarrow \mathbf{KleisliCat} \ \mathbf{PMF} \ (\mathbf{InfoFunctor.lean})$

Object $n \mapsto \mathbf{Fin} \ n$, the type of n distinguishable outcomes.

Morphism ($n \leq m$): the deterministic embedding (each outcome mapped to its image with certainty).

$\mathbf{fC_zero_isInitial}$: $\mathbf{Fin} \ 0$ (the empty type) is the initial object.

$\mathbf{fC_no_return}$: for $n > 0$ there is NO stochastic map $\mathbf{Fin} \ n \rightarrow \mathbf{Fin} \ 0$.

AX-G2 as a theorem: no probability distribution lives on the empty type, so nothing returns to the floor.

Sorry-free. Purity [propext, Classical.choice, Quot.sound].

The empty type is the snap floor here, and it carries the snap's irreversibility for free. A stochastic map into $\mathbf{Fin} \ 0$ would be a probability distribution on the empty type, and there is none. So while every object has a unique map out of $\mathbf{Fin} \ 0$ (it is initial), no nonempty object has any map back — AX-G2 (source asymmetry) is not assumed but proved, as $\mathbf{fC_no_return}$. Of the three native categories, this is the one where the framework's irreversibility axiom becomes a theorem of the ambient mathematics.

Section V: The MC-1 Correspondence, Scope, and Purity

The three functors, together with $\mathbf{F_A}$ (the join-semilattice $\mathbb{N}$ of the main document, where 0 is already the initial object), are bundled as a single witness.

Definition: $\mathbf{mc1_correspondence} : \mathbf{MC1Correspondence} \ (\mathbf{MC1Bridge.lean})$

A record collecting the native-category realizations:

- $\mathbf{F_D}$: $\mathbf{StateSpace} \ 0$ is initial in $\mathbf{ModuleCat} \ \mathbb{C}$.

Definition: `mc1_correspondence` : `MC1Correspondence` (`MC1Bridge.lean`)

- `F_C`: `Fin 0` is initial in `KleisliCat` PMF, with no return morphism.

- `F_B`: the clopen-ball tower has inverse limit $\{0\}$ in `TopCat`.

Sorry-free. Purity [`propext`, `Classical.choice`, `Quot.sound`].

This settles the correspondence half of MC-1: across the native categories, the snap floor is each category's own categorical bottom, and the four agree on the snap. The identity half, that the algebraic $\perp$, the 2-adic 0, the zero module, and the empty type are numerically one object, is retired as ill-typed (object equality across categories does not typecheck and is not invariant under equivalence); what separates the members is proved property by property (`seam_unique_among_named`, for the named bottoms, in a lattice with no top).

Scope and Purity

This is a formalisation milestone, not new category theory. `TopCat`, `ModuleCat` $\mathbb{C}$, and the Kleisli category of PMF are standard Mathlib categories with standard universal properties. The contribution is realizing the ZP snap floor inside each as a genuine functor, in place of the $\mathbb{N}$ -indexed depth proxies of the main document.

Purity: [`propext`, `Classical.choice`, `Quot.sound`]. Unlike the choice-free ZP-J results, these functors inherit `Classical.choice` from Mathlib's topology, inner-product, and PMF libraries. The dependency is a library inheritance, not a new commitment of the construction; whether it is structurally forced is the open question tracked under `Classical.choice` necessity (ZP-L / ZP-M).

Lean Source Files

`TopFunctor.lean` — `fB_functor`, `fB_bottom_is_limit` (`F_B` into `TopCat`).

`HilbFunctor.lean` — `fD_functor`, `fD_zero_isInitial`, `fD_embed_inner` (`F_D` into `ModuleCat` $\mathbb{C}$).

`InfoFunctor.lean` — `fC_functor`, `fC_zero_isInitial`, `fC_no_return` (`F_C` into `KleisliCat` PMF).

`MC1Bridge.lean` — `mc1_correspondence` (the bundled witness).

All in `ZeroParadox/` in the public repository.

Endnote: This document is an addendum to ZP-H Categorical Bridge and reads after it. ZP-H locates the snap floor as the categorical bottom across four frameworks using depth-index proxy categories; this addendum realizes the same floor inside each framework's native Mathlib category — `TopCat`, `ModuleCat` $\mathbb{C}$, and the Kleisli category of distributions.

All results sorry-free in Lean 4 as of June 2026, footprint [`propext`, `Classical.choice`, `Quot.sound`].